

Arithmetic and Order of Real Numbers

- **Definition 1**

The set of **real numbers**, denoted $\mathbb{R}$, is a set that contains $\mathbb{Q}$. Its sum, product, and order extend the operations and order in $\mathbb{Z}$ and satisfy the following properties for all $x, y, z \in \mathbb{R}$:

1.
$$(x + y) + z = x + (y + z).$$

2. There exists a unique element $0 \in \mathbb{R}$ such that

$$x + 0 = x.$$

3. For each $x \in \mathbb{R}$, there exists a unique $-x \in \mathbb{R}$ such that

$$x + (-x) = 0.$$

4.
$$x + y = y + x.$$

5.
$$(xy)z = x(yz).$$

6. There exists a unique element $1 \in \mathbb{R}$ such that

$$x \cdot 1 = x.$$

7. For each $x \in \mathbb{R}$ with $x \neq 0$, there exists a unique element $x^{-1} \in \mathbb{R}$ such that

$$x \cdot x^{-1} = 1.$$

8.
$$xy = yx.$$

9.
$$x(y + z) = xy + xz.$$

10. The relation $\leq$ is a total order in $\mathbb{R}$.

11. If $x \leq y$, then $x + z \leq y + z$.

12. If $x \leq y \wedge z \geq 0$, then $xz \leq yz$.

- **Definition 2**

The set of **irrational numbers** is the set of those real numbers that are not rational numbers, that is,

$$\mathbb{R} \setminus \mathbb{Q}.$$

- **Example 2.1**

$$\sqrt{2}, e, \pi.$$

- **Proposition 2.2**

The number $\sqrt{2}$ is irrational.

- **Proof 2.2.1**

Suppose that $\sqrt{2} = \frac{p}{q}$ with $p \in \mathbb{Z}$, $q \in \mathbb{N}$ and p, q are coprime. Then

$$p^2 = 2q^2.$$

Hence $2 \mid p^2$, and since 2 is a prime we have that $2 \mid p$, that is, there exists $k \in \mathbb{Z}$ such that $p = 2k$. So

$$4k^2 = 2q^2,$$

$$2k^2 = q^2.$$

Then $2 \mid q^2$, and since 2 is a prime we have that $2 \mid q$, which is a contradiction since p, q are coprime.

◦ **Proposition 2.3**

Let $a, b, c, d \in \mathbb{R}$, then:

1. If $a < b$ and $c < 0$, then $ac > bc$.
2. If $a < b$ and $c < d$, then $a + c < b + d$.
3. If $a < b$, then $-a > -b$.
4. If $a < 0$ and $b < 0$, then $ab > 0$.

▪ **Proof 2.3.1**

1. Since $-c > 0$ and $b - a > 0$, then

$$ac - bc = (b - a)(-c) > 0.$$

And thus $ac > bc$.

2. Since $b - a > 0$ and $d - c > 0$, then

$$(b + d) - (a + c) = (b - a) + (d - c) > 0.$$

And thus $a + c < b + d$.

3. By Proposition 2.3(1) with $c = -1$.

4. Since $-a > 0$ and $-b > 0$, then $ab = (-a)(-b) > 0$.

◦ **Proposition 2.4**

$$a > 0 \iff \frac{1}{a} > 0.$$

▪ **Proof 2.4.1**

($\implies$) Suppose $a > 0$. By the axiom of the multiplicative inverse, there exists $a^{-1} = \frac{1}{a}$ such that $a \cdot \frac{1}{a} = 1$. Since $1 > 0$ and $a > 0$, it follows from the properties of order and multiplication that $\frac{1}{a}$ must be positive, as a positive number multiplied by a negative number would yield a negative result. Thus, $\frac{1}{a} > 0$.

($\impliedby$) Suppose $a \in \mathbb{R}$ such that $\frac{1}{a} > 0$. Applying the forward implication ($\implies$) to the positive real number $\frac{1}{a}$, we have that its multiplicative inverse must also be positive, that is, $\frac{1}{\frac{1}{a}} > 0$. Since $a \cdot \frac{1}{a} = 1$, multiplying both sides by $\frac{1}{\frac{1}{a}}$ yields $a = \frac{1}{\frac{1}{a}}$. Therefore, we have $a > 0$.

• **Definition 3**

Let $x \in \mathbb{R}$ and $n \in \mathbb{N}$, we define

$$x^n = \underbrace{x \cdot x \cdots x}_{n \text{ times}}.$$

If $x \neq 0$ we define $x^0 = 1$ and $x^{-n} = (x^{-1})^n$.

◦ **Theorem 3.1**

Let $x, y \in \mathbb{R}$ and $m, n \in \mathbb{N}$, then

1. $x^{m+n} = x^m \cdot x^n$.
2. $(x^m)^n = x^{mn}$.
3. $(xy)^n = x^n y^n$.
4. $\frac{x^m}{x^n} = x^{m-n}$ if $x \neq 0$.
5. $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$ if $y \neq 0$.

■ **Proof 3.1.1**

1. Fix $m \in \mathbb{N}$ and use induction on n .

For $n = 1$ the result is true since $x^{m+1} = x^m x = x^m x^1$.

Suppose that the result is true for n , that is

$$x^{m+n} = x^m x^n.$$

Hence

$$\begin{aligned} x^{m+(n+1)} &= x^{(m+n)+1} \\ &= x^{m+n} x \\ &= (x^m x^n) x \\ &= (x^m) (x^n x) \\ &= x^m x^{n+1}. \end{aligned}$$

Thus the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

2. Fix $m \in \mathbb{N}$ and use induction on n .

For $n=1$ the result is true since $(x^m)^1 = x^m = x^{m \cdot 1}$.

Suppose that the result is true for n , that is

$$(x^m)^n = x^{mn}.$$

Hence

$$\begin{aligned} (x^m)^{n+1} &= (x^m)^n \cdot x^m \\ &= x^{mn} \cdot x^m \\ &= x^{mn+m} \\ &= x^{m(n+1)}. \end{aligned}$$

Thus the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

3. Induction on n .

For $n = 1$ the result is true since $(xy)^1 = xy = x^1 y^1$.

Suppose that the result is true for n , that is

$$(xy)^n = x^n y^n.$$

Hence

$$\begin{aligned} (xy)^{n+1} &= (xy)^n (xy) \\ &= x^n y^n (xy) \\ &= (x^n x) (y^n y) \\ &= x^{n+1} y^{n+1}. \end{aligned}$$

Therefore the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

4. Fix $m \in \mathbb{N}$ and suppose $x \neq 0$. Use induction on n .

For $n = 1$ the result is true since $\frac{x^m}{x^1} = x^m x^{-1} = x^{m+(-1)} = x^{m-1}$ by Property 1.

Suppose that the result is true for n , that is

$$\frac{x^m}{x^n} = x^{m-n}.$$

Hence

$$\begin{aligned} \frac{x^m}{x^{n+1}} &= \frac{x^m}{x^n \cdot x} \\ &= \frac{x^m}{x^n} \cdot x^{-1} \\ &= x^{m-n} \cdot x^{-1} \\ &= x^{(m-n)+(-1)} \\ &= x^{m-(n+1)}. \end{aligned}$$

Thus the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

5. Suppose $y \neq 0$. Use induction on n .

For $n = 1$ the result is true since $\left(\frac{x}{y}\right)^1 = \frac{x}{y} = \frac{x^1}{y^1}$.

Suppose that the result is true for n , that is

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}.$$

Hence

$$\begin{aligned} \left(\frac{x}{y}\right)^{n+1} &= \left(\frac{x}{y}\right)^n \cdot \left(\frac{x}{y}\right) \\ &= \frac{x^n}{y^n} \cdot \left(\frac{x}{y}\right) \\ &= \frac{x^n \cdot x}{y^n \cdot y} \\ &= \frac{x^{n+1}}{y^{n+1}}. \end{aligned}$$

Thus the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

◦ **Proposition 3.2**

If $n \in \mathbb{N}$ and $0 \leq x < y$, then $x^n < y^n$. In particular, if $x, y \geq 0$ and $x^n = y^n$, then $x = y$.

■ **Proof 3.2.1**

Use induction on n .

If $n = 1$, then the result is true since $x^1 < y^1$.

Suppose that the result is true for n , that is,

$$x^n < y^n.$$

Hence,

$$x^{n+1} = x^n x \leq x^n y < y^n y = y^{n+1}.$$

Thus the result is true for $n + 1$. So the result is true for every $n \in \mathbb{N}$.

Suppose that $x, y \geq 0$, $x^n = y^n$ and $x \neq y$.

If $x < y$, then $x^n < y^n$, which is a contradiction.

If $x > y$, then $x^n > y^n$, which is a contradiction.

Therefore $x = y$.

- **Definition 4**

If $x \geq 0$ and $n \in \mathbb{N}$, the **n-th root** of x , denoted $\sqrt[n]{x}$, is the unique nonnegative number y such that $y^n = x$.

- **Example 4.1**

The equation $y^2 = 9$ has two solutions: -3 and 3 . But $\sqrt{9} = 3$ denotes only the nonnegative one.

- **Remark 4.2**

A negative real number has no even root, whereas an odd root is defined by

$$\sqrt[n]{x} = -\sqrt[n]{-x}$$

if $x < 0$ and n is odd.

If $x^2 < y^2$, must $x < y$? No. For example, $(-1)^2 < (-2)^2$ but $-1 > -2$. The implication is true when both numbers are nonnegative.

If $x, y \geq 0$ with $x^n < y^n$, then $x < y$. By Proposition 3.2, since if $x = y$, then $x^n = y^n$, and if $x > y$, then $x^n > y^n$, which is a contradiction.